

PhysInOne: Visual Physics Learning and Reasoning in One Suite

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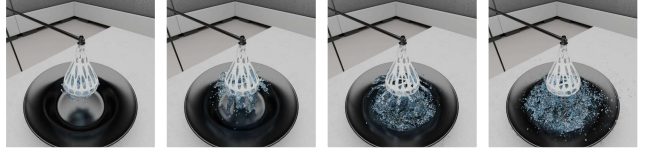
<https://vlar-group.github.io/PhysInOne.html>

MECHANICS



A car accelerated ... wooden box wall ... collision displaced the box connected to the green balloon ... buoyant force ... began rising ...

FLUID DYNAMICS



A triangular vase ... rests atop a dish ... Liquid within the vase ... drawn downward by gravity ... The droplets splatter gently upon ...

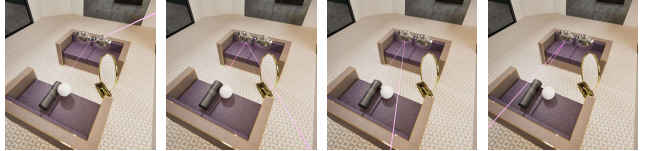


MAGNETISM



... wooden seesaw rested horizontally ... A red magnet occupied ... a metal block ... magnetic attraction ... created an imbalance ...

OPTICS



... laser pen emitted light toward a convex mirror ... ball rolled across the sofa ... collided with ... altered the incident angle ...

Figure 1. We present PhysInOne, a large-scale dataset of 153,810 dynamic 3D scenes and 2 million annotated videos, systematically capturing 71 basic physical phenomena and multiobject interactions to advance visual physics learning and reasoning.

Abstract

We present **PhysInOne**, a large-scale synthetic dataset addressing the critical scarcity of physically-grounded train-

ing data for AI systems. Unlike existing datasets limited to merely hundreds or thousands of examples, **PhysInOne** provides 2 million videos across 153,810 dynamic 3D scenes,

covering 71 basic physical phenomena in mechanics, optics, fluid dynamics, and magnetism. Distinct from previous works, our scenes feature multiobject interactions against complex backgrounds, with comprehensive ground-truth annotations including 3D geometry, semantics, dynamic motion, physical properties, and text descriptions. We demonstrate PhysInOne’s efficacy across four emerging applications: physics-aware video generation, long-/short-term future frame prediction, physical property estimation, and motion transfer. Experiments show that fine-tuning foundation models on PhysInOne significantly enhances physical plausibility, while also exposing critical gaps in modeling complex physical dynamics and estimating intrinsic properties. As the largest dataset of its kind, orders of magnitude beyond prior works, PhysInOne establishes a new benchmark for advancing physics-grounded world models in generation, simulation, and embodied AI.

1. Introduction

Large-scale, high-quality data serves as the catalyst for modern machine learning breakthroughs. In natural language processing, massive corpora like Common Crawl [21] have enabled the unprecedented abilities of LLMs such as GPT-4 [64]. A similar paradigm has driven 2D computer vision: foundational datasets like ImageNet [25] and MS-COCO [57] accelerated discriminative tasks, while massive-scale collections like LAION-5B [73] have revolutionized image generation. Extending this to the 3D/4D domain, vast repositories such as Objaverse-XL [23, 24] have facilitated rapid advances in reconstruction, while Panda-70M [17] has become a cornerstone for video generation and world models.

Despite impressive results demonstrated by state-of-the-art LLM/LVM/MLLM/GenAI models trained on these datasets, as shown in [47, 63], their ability to truly comprehend the physical world and its laws remains limited, a challenge that has long been regarded as a major milestone on the path toward general-purpose AI. For example, while visually appealing, videos generated by AI frequently violate basic physical laws, with objects falling upward or suddenly shifting velocity. This has spurred growing interest in learning physics from visual and/or texts in recent works [5, 14, 33, 34, 46, 52–55, 58, 59, 94, 96, 101, 103]. Despite achieving encouraging results, these models are typically trained on only dozens, hundreds, or thousands of data points, which restricts them to limited physical phenomena, making them far from true world simulators. This limitation primarily stems from the lack of large-scale, high-quality training data that capture various physical objects and scenes, and encompass diverse physical phenomena in 3D space.

In this paper, we present **PhysInOne**, a large-scale synthetic dataset featuring richly annotated dynamic 3D scenes. These scenes depict a wide range of physical phenomena among multiple objects interacting against complex back-

grounds. Drawing on the introductory university-level textbook *Fundamentals of Physics* [39] and recent research [62, 63], we focus on visual learning in four core areas of everyday physics: mechanics, optics, fluid dynamics, and magnetism, while omitting thermodynamics and acoustics, which are rarely visible or require additional sensory data like temperature or sound beyond the scope of our dataset.

Across these four areas, we identify 71 key physical phenomena commonly encountered in daily life, such as gravity, reflection, buoyancy, magnetic attraction, *etc.*, covering the vast majority, if not all, of relevant everyday physics. For these phenomena, we create 153,810 dynamic 3D scenes. Each scene consists of multiple objects and undergoes multiple physical phenomena simultaneously or sequentially, all strictly governed by fundamental physical laws including Newton’s Laws, Mass Conservation, (Angular) Momentum Conservation, Hooke’s Law, *etc.*. For each dynamic 3D scene, we record 13 videos capturing the entire physical activity: 12 from fixed cameras and 1 from a smoothly moving monocular camera. Alongside these videos, we manually annotate a high-quality paragraph of text, accurately describing the scene’s visual elements and activity. This yields a comprehensive dataset of 2 million dynamic videos with ground truth annotations, including object 3D meshes and moving trajectories, 2D masks, materials, depths, text descriptions, camera poses, *etc.*, being orders of magnitude larger than all existing visual physics datasets.

With this significant increase in scale and diversity, we believe PhysInOne has the potential to make a substantial impact on research progress in visual physics learning. In this paper, we present four emerging application examples by exploiting our dataset, including:

1) Physics-aware Video Generation: Video generation is often touted as a promising path towards simulating the physical world [65], but currently still lacks the essential ability to predict physically plausible clips. By fine-tuning SVD [10], CogVideoX [91], and WAN [83] models using just a moderate subset of text-video pairs from PhysInOne, we significantly boost the physical plausibility of generated videos. This improvement is evidenced by our newly introduced metric focused on physical motion fidelity.

2) (Long-/Short-term) Future Frame Prediction: As physicist Richard Feynman stated, “What I cannot create, I do not understand”. This insight implies that predicting how videos continue is a crucial task to test a model’s understanding of physical principles. This ability also serves as a key enabler for emerging applications in future-aware robot planning and embodied AI [26]. To this end, we showcase this critical task on PhysInOne by training and evaluating a series of recent models, including TiNeuVox [31], DefGS [90], FreeGave [54], TRACE [53], ExtDM [98], and MAGI-1 [1].

3) Physical Properties Estimation: Inferring physical properties from visual observations, a.k.a system identifi-

Table 1. A comparison between PhysInOne and existing datasets that are relevant to learning physics from dynamic videos. PhysInOne provides massive 3D scenes with complex objects and backgrounds across diverse physical phenomena, surpassing all prior works.

	Year	# Physical Phenomena	# Physical Scenes	# Dynamic Videos	Multiobject Interactions	Multiphysics Activities	Complex Objects	Complex Backgrounds	# Fixed / Monocular Cameras	Annotations (Geometry, Semantics, Motion, Properties, Texts)
Physics101 [88]	BMVC'16	4	-	17,408	✗	✗	✗	✗	3, 0	P
ShapeStacks [36]	ECCV'18	1	20,000	20,000	✓	✗	✗	✗	1, 0	S, M, T
ScalarFlow [29]	TOG'19	1	100	100	✗	✗	✗	✗	- , -	M, P
CLEVRER [92]	ICLR'20	1	10,000	10,000	✓	✗	✗	✗	- , -	M, T
Physion [9]	NeurIPS'21	8	-	24,000	✓	✗	✗	✓	- , -	M, T
ComPhy [18]	ICLR'22	1	12,000	12,000	✓	✗	✗	✗	1, 0	M, T
SPACE+ [28]	ECCV'22	3	15,000	57,057	✓	✗	✗	✗	1, 0	G, S, M
Physion++ [80]	NeurIPS'23	9	-	~ 18,000	✓	✗	✗	✗	- , -	P, T
PAC-NeRF [55]	NeurIPS'23	-	9	99	✗	✗	✗	✗	11, 0	P
VideoPhy [6]	ICLR'25	-	-	11,330	✓	-	✓	✓	0, 1	T
PhysTwin [45]	ICCV'25	-	22	66	✓	✗	✓	✗	3, 0	M, P
IntPhys2 [12]	arXiv'25	-	344	1416	✓	✗	✓	✓	- , -	T
Physics-IQ [63]	arXiv'25	-	66	396	✓	✗	✓	✗	3, 0	M
PhysVid [96]	arXiv'25	3	-	~33,000	✗	✗	✗	✗	- , -	S, M, P
NewtonGen [94]	ICLR'26	12	-	1200	✗	✗	✓	✓	1 , -	T
PhysInOne (Ours)	CVPR'26	71	153,810	2 million	✓	✓	✓	✓	12, 1	G, S, M, P, T

cation or inverse physics, aims to learn disentangled, interpretable, and editable physical quantities, thus enabling various applications like resimulation, scene editing, robot manipulation, *etc.* We benchmark this task on our dataset using two recent representative methods PAC-NeRF [55] and GIC [14] to estimate a variety of parameters like Young’s modulus, Poisson’s ratio, viscosity, *etc.*

4) Motion Transfer: Transferring motions between scenes enables entertaining image animations and controllable generation. By evaluating recent methods MotionPro [99] and GoWithTheFlow [13] on PhysInOne, we showcase the pipeline of transferring physically meaningful motions from one scene to another. However, the results remain largely unsatisfactory, underscoring the unique challenge in transferring complex physical motion patterns.

Overall, these findings, enabled by our large-scale, high-quality PhysInOne dataset, represent just a fraction of its potential. We anticipate the research community will leverage it to accelerate progress in 3D/4D vision and beyond.

2. Related Datasets and Benchmarks

Datasets for Learning Physical Dynamics: Existing datasets for learning and reasoning physical dynamics often focus on a narrow set of physical phenomena. For instance, ShapeStacks [36] and StableText2Brick [68] datasets only address object or part stability, while datasets like VOE [66], IntPhys2019 [72], PHYRE [4], ADEPT [78], CLEVRER [92], ESPRIT [70], Physion [9], CoPhy [8], CRAFT [2], and SPACE+ [28] examine simple scenarios where single object trajectory follows intuitive physics such as object persistence/continuity/solidity, rather than rigorously verifying adherence to a wide range of strict physical laws. In addition, almost all objects in these datasets are oversimplified shapes (cubes, balls, *etc.*) with homogeneous colors against clean backgrounds. Other datasets specialize in learning fluid dynamics from visual data: ScalarFlow [29], TomoFluid

[95], and FLAME [74] focus on smoke, while Obstacles [3], Richter *et al.* [71], and NeuroFluid [38] on liquids.

Datasets for Learning Physical Properties: A growing number of datasets have been collected for estimating different physical properties [86–88], including Physics101 [88] for mass and volume, Physion++ [80], Materialistic [75], PAC-NeRF [55], PhysTwin [45], PhysXNet [16], SO-PHY [15], PixieVerse [48], PhysVid [96], and VoMP [22], for various physical properties like mass, friction, elasticity, deformability, *etc.* These datasets usually focus on isolated individual objects or oversimplified 3D scenes.

Datasets for Testing Physical Understanding: Recent advances of large language, vision, multimodal, video, and world models have led to a series of benchmarking datasets to test the physical understanding abilities of these models, including ComPhy [18], PerceptionTest [69], TraySim [19], GRASP [44], PhyBench [61], Physics-IQ [63], Morpheus [97], WorldModelBench [51], WISA [84], VBench-2.0 [102], PhysBench [20], DynSuperCLEVR [85], VideoPhy [6], VideoPhy-2 [7], PhyX [76], IntPhys2 [12], PhyGenBench [62], UGPhysics [89], STI-Bench [56], PhyWorldBench [37], PisaBench [50], NewtonGen [94], and NewtonBench-60K [49]. These benchmarks typically focus on text–image–video question answering (QA) or visual understanding and generation in narrow scenarios, and thus lack the precise and diverse visual data and annotations needed to train or fine-tune large, general models for quantitatively evaluating visual physics learning.

Table 1 compares PhysInOne with key existing datasets designed for physics learning from videos, highlighting our scale and diversity across multiple dimensions.

3. PhysInOne

The pipeline to construct our dataset involves multiple stages as detailed in the following sections and shown in Figure 2.

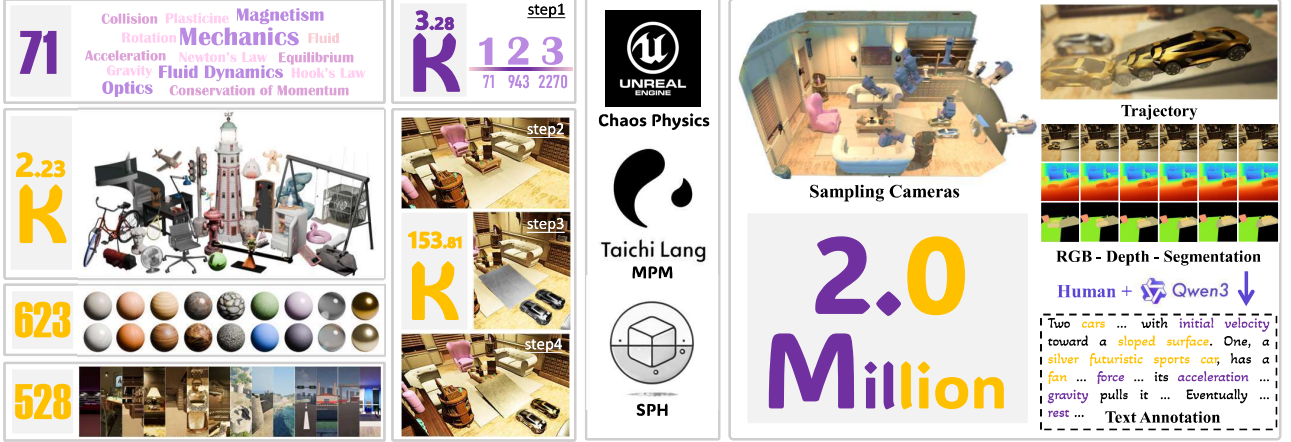


Figure 2. Spanning 71 basic physical phenomena scaled to 3,284 multiphysics activities, PhysInOne comprises 153,810 unique scenes featuring 2,231 objects across 528 backgrounds and 623 materials, yielding 2 million videos with comprehensive annotations.

3.1. Physical Phenomena and Laws

Within the four core areas of everyday physics, mechanics, optics, fluid dynamics, and magnetism, we identify 71 basic physical phenomena, covering the predominant physics commonly observed in daily life. Each basic phenomenon is governed by one or more fundamental physical laws, such as Newton’s Laws, Conservation of Momentum, Hooke’s Law, Bernoulli’s Principle, *etc.*. A complete list of these basic phenomena and their governing laws is in Appendix 5.1.

Importantly, each basic physical phenomenon represents a conceptual occurrence in daily life, not a specific, concrete 3D scene. In Section 3.3, we will create tangible 3D scenes that instantiate one or more of these basic phenomena.

3.2. Collecting 3D Assets

In real-world scenarios, a physical 3D scene typically comprises multiple objects interacting with each other against complex backgrounds. The dynamics of such scenes instantiate one or more basic physical phenomena and are strictly governed by the fundamental physical laws identified in Section 3.1. To replicate as many diverse examples of these complex 3D scenes as possible, we collect a rich set of 3D assets for use in Section 3.3. These assets include:

- **3D Objects:** Since our dataset targets everyday physics and particularly the identified 71 basic physical phenomena, we collect 2,231 common objects (~ 163 categories) from Sketchfab [77], FAB [30], and BlenderKit [11]. These objects, directly relevant to physical interactions, distinguish our dataset from those like Objaverse/XL designed for generic 3D tasks. The collection includes:
 - *solid objects*: like hammers and bricks that maintain their shape during physical interactions.
 - *interactable objects*: which have movable parts (*e.g.*, rotatable fans), implemented as Blueprints in UE [81].
 - *destructible objects*: such as breakable glasses, imple-

mented as Geometry Collections in UE.

- *deformable objects*: whose shape changes during physical activities. Their deformation is simulated using MPM [43], as detailed in Section 3.4.
- *granular objects*: primarily representing granular materials like sand, also simulated using MPM.
- *liquid*: like water, cream, milk, *etc.*, simulated using SPH [35], as detailed in Section 3.4.
- **Materials:** To achieve greater realism and diversity in object representation, we further collect 623 materials across 5 categories: plastic, metal, wood, stone, and fabric. Each material’s adjustable physical properties, such as friction coefficients, density, restitution, enables the creation of numerous scene variations, as discussed in Section 3.3.
- **3D Backgrounds:** To provide realistic environments where diverse physical phenomena occur, we collect 528 distinct 3D backgrounds. These encompass various indoor and outdoor scenarios at different spatial scales, such as living rooms, bedrooms, factories, swimming pools, *etc.*. Notably, all collected 3D assets are freely available and licensed for both academic and commercial use. More details of 3D assets are provided in Appendix 5.2.

3.3. Creating Multiphysics Multiobject 3D Scenes

With the collected 3D assets and identified 71 basic physical phenomena, we create massive, realistic, and challenging 3D scenes through the following steps:

- **Step 1: Emulating Multiphysics:** Since daily physical activities often involve one or more basic physical phenomena simultaneously or sequentially, we emulate 3 categories:
 - *single-physics activities*: We treat each of the 71 basic phenomena as a distinct single-physics activity.
 - *double-physics activities*: We combine 2 basic phenomena into a complex activity. While C_{71}^2 combinations exist, many lack physical meaning. After careful selec-

tion, we create 943 double-physics activities.

- *triple-physics activities*: Similarly, we combine 3 basic phenomena into a more complex activity. After selection, we create 2270 valid triple-physics activities.

The resulting $71 + 943 + 2270 = 3284$ physical activities are still conceptual at this stage; they are implemented as tangible 3D scenes in subsequent steps.

- *Step 2: Setting Up Backgrounds*: We set up physically plausible backgrounds aligned with the target activity.
- *Step 3: Placing Multiobjects*: For each activity, we create many tangible 3D scenes. In each scene, we select one or more 3D assets and place them into a background.
- *Step 4: Varying Materials*: For 3D assets in each scene, we alter their materials and vary physical properties.

Following the above four steps, we create 153,810 unique multiphysics multiobject 3D scenes. Each activity is realized through an average of 46.84 distinct scenes featuring varied assets, materials, and backgrounds. The average number of 3D assets (objects) per scene increases with activity complexity: 3.9/6.3/7.8 for single-/double-/triple- physics activities, demonstrating the escalating scene complexity. More details are in Appendix 5.3.

3.4. Simulating Physical Dynamics

To simulate physical activities of our 153,810 scenes as accurate as possible, we employ multiple simulation algorithms, packages and engines. In particular, *Chaos Physics* integrated in UE5 [81] handles the majority of daily physical phenomena. For deformable and granular objects, we utilize MPM implemented by Taichi [79]. Liquid simulations employ SPH through Doriflow [27]. More details of simulation, including the ranges of various physical parameters, are provided in Appendix 5.4.

We note that while current simulators may not achieve perfect physical fidelity, their errors are thoroughly studied and controlled during simulation [32, 60]. Thus, creating a massive synthetic dataset encompassing diverse physical phenomena unarguably holds significant potential to advance research progress, as also demonstrated by recent work [100].

3.5. Sampling Cameras and Rendering Videos

For each 3D scene, we place 12 static cameras uniformly across the upper hemisphere at elevation angles $30^\circ \sim 60^\circ$ to capture comprehensive multiview coverage, while a single moving camera records a challenging monocular video throughout the entire physical activity. Note that, the static cameras’ positions vary across scenes according to different spatial arrangements and backgrounds. The moving camera follows predefined trajectories, typically circling the activity at random elevation angles. All videos are rendered at 1120×1120 resolution with 30 FPS (60 FPS for 8780 scenes containing laser activities), averaging 5.2 seconds in dura-

tion to fully capture each physical activity. This results in 2 million dynamic videos. More details are in Appendix 5.5.

3.6. Annotating and Splitting Dataset

For each 3D scene, we provide comprehensive annotations including a detailed paragraph describing visual elements and the physical activity. These texts are manually added and proofread using Qwen3 to eliminate grammatical errors, averaging about 64 English words per scene. During video rendering, we simultaneously generate: ground truth depth images, per-frame object masks, 3D trajectories of dynamic objects, object meshes, and material properties. Our annotations thus encompass five key aspects: geometry, semantics, motion, physical properties, and textual descriptions.

The dataset is partitioned into train/validation/held-out test sets using an 8:1:1 ratio. We ensure all 3D assets appear exclusively in one partition to prevent potential data leakage. Our dataset quality is ensured through a structured workflow including standardized development and independent validation. More details are in Appendix 5.6.

4. Applications

4.1. Physics-aware Video Generation

While video generation models demonstrate significant advancements in visual fidelity, they frequently fail to capture physically plausible dynamics [47, 63]. PhysInOne contains diverse and massive dynamic activities following a variety of daily physical phenomena. This rich repository serves as an ideal resource for training or fine-tuning next-generation video models that faithfully emulate real-world physics.

In this section, we conduct fine-tuning experiments on three representative video models: 1) **SVD-XT** [10], a UNet-based image-to-video (I2V) model; 2) **CogVideoX-1.5-5B** [91], a Transformer-based text-image-to-video (TI2V) model; and 3) **Wan2.2-5B** [83], the latest Transformer and flow matching based TI2V model in the field. For these models, we adopt three commonly used fine-tuning techniques: 1) Low-Rank Adaptation (**LoRA**) [42], 2) Supervised Fine-Tuning (**SFT**) [41], and 3) Final Layer Tuning (**FLT**) [93]. To showcase the potential of PhysInOne, we randomly sample a subset of training videos (83,650 text-video pairs) for fine-tuning all models until convergence. All fine-tuned models and their original models are then evaluated on a subset of test videos (772 text-video pairs, called *test-small*). More details of experiment settings are provided in Appendix 5.7.

Metrics: Traditional metrics such as **FVD** [82] primarily assess visual realism in video generation, but are inadequate for evaluating the physical plausibility of motion. Some recent studies [6, 40] use video-based VLMs to assess physical commonsense. However, these models often fail to produce meaningful evaluations, as they fundamentally lack an understanding of physical laws and are thus ill-suited

for judging physical correctness. Other works [20, 62, 76] introduce benchmarks featuring QA tasks to probe physical understanding, but they are typically qualitative and cannot quantitatively measure the correctness of physical motions.

To this end, we introduce a novel metric for the quantitative assessment of physical motion fidelity in generated videos. Particularly, given a reference video $\mathcal{V}_{ref}$ exhibiting physically accurate motion (*e.g.*, from our test set) and an AI-generated video $\mathcal{V}_{gen}$ produced using identical initial frame(s) and textual prompts as $\mathcal{V}_{ref}$, ensuring controlled comparison conditions, we apply discrete Fourier transform (DFT) to obtain their respective frequency domain representations. We then compare the energy (squared amplitude) of $DFT(\mathcal{V}_{ref})$ and $DFT(\mathcal{V}_{gen})$ and introduce **Physical Motion Fidelity (PMF)**, defined as follows:

$$PMF = f_{energy}(DFT(\mathcal{V}_{gen}), DFT(\mathcal{V}_{ref})) \quad (1)$$

PMF quantifies kinematic discrepancies between the dynamic trajectories in $\mathcal{V}_{gen}$ and $\mathcal{V}_{ref}$. Crucially, higher PMF scores indicate less deviation from the reference motion patterns. The metric fundamentally differs from pixel-level similarity measures by evaluating physical fidelity rather than visual correspondence, as frame-perfect alignment is neither achievable nor desirable in generative tasks.

To empirically validate physical plausibility from a human perspective, we additionally conduct a user study to assess generated videos. Higher human ratings reflect greater perceived physical plausibility in the video content. More details of our metric and human rating are in Appendix 5.7.

Table 2. The left part shows quantitative results of video generation models with and without fine-tuning on PhysInOne. The right part shows PMF scores across four physical domains separately.

	PMF $\uparrow$	FVD $\downarrow$	Human Rating $\uparrow$	Mechanics PMF $\uparrow$	Magnetism PMF $\uparrow$	Optics PMF $\uparrow$	Fluid PMF $\uparrow$
SVD [10]	2.753	203	6.09	2.763	3.758	2.582	3.278
SVD _{lora}	2.446	150	5.82	2.473	3.283	2.303	2.815
SVD _{sft}	3.147	143	6.08	3.139	3.948	3.261	3.868
SVD _{flt}	2.464	147	5.45	2.463	3.426	2.485	3.061
CogVideoX [91]	2.877	165	2.98	2.912	3.025	2.495	3.033
CogVideoX _{lora}	2.869	149	2.95	2.881	3.008	2.482	3.206
Wan2.2-5B [83]	2.041	258	2.26	2.031	2.752	1.588	3.205
Wan2.2-5B _{lora}	2.785	178	4.80	2.760	3.390	2.566	3.716
Wan2.2-5B _{sft}	2.978	190	5.95	2.990	3.462	2.888	3.344
Wan2.2-5B _{flt}	2.227	341	2.61	2.227	3.214	1.908	3.325

The left part of Table 2 shows that SVD and Wan2.2 achieve notable improvements in PMF scores after fine-tuning, especially using SFT technique. This validates the effectiveness of PhysInOne in imbuing models with physical knowledge. We also observe that human perception of physical dynamics highly correlates with our newly introduced PMF metric. The right part of Table 2 shows that most models exhibit higher accuracy in magnetism and fluid, but lower scores in mechanics and optics, highlighting the challenges for future research in learning different types of physics. We will continue to benchmark the latest video generation models and update results on our [website](#).

4.2. Future Frame Prediction

Accurate prediction of future frames necessitates a model’s comprehension of underlying physical dynamics, with critical applications in autonomous driving, embodied AI, *etc.*. Input-output specifications may vary significantly in different applications. For example, in robot manipulation, given multiview videos as input, the model aims to predict precise dexterous control, where accurate and continuous short-term future predictions are essential for action execution. Conversely, video understanding tasks typically prioritize long-term predictions from monocular video inputs. Our collection of both multiview and monocular video data in PhysInOne enables demonstration of these diverse applications. This section introduces two tasks: long-term and short-term future frame prediction under varied settings.

4.2.1. Long-term Future Frame Prediction

In this experiment, models are tasked with predicting the second half of a dynamic 3D scene (~ 2.6 seconds ahead, ~ 78 future frames) from the test set, given the first half video clip as input. We evaluate the following two groups of existing methods for this task.

- **4D Modeling Methods:** **TiNeuVox** [31], **DefGS** [90], **FreeGave** [54], and **TRACE** [53]. These methods model dynamic 3D scenes from multiview videos using scene-specific deformation or velocity fields, enabling future frame prediction from arbitrary (seen and novel) views.
- **Video Prediction Methods:** **ExtDM** [98], and **MAGI-1** [1]. ExtDM is trained from scratch on randomly sampled 83,650 videos (same as used in Section 4.1), while MAGI-1 is evaluated using its pretrained model.

Given that training scene-specific models for the entire test set is neither feasible nor necessary, we evaluate all six methods on a randomly selected subset of 103 test scenes, called *test-mini*. This assesses their long-term future prediction capabilities for both seen and novel viewing angles. We report standard per-frame metrics **PSNR/SSIM/LPIPS**, along with our proposed dynamics metric, **PMF**. More details of experiment settings are provided in Appendix 5.8.

Table 3. Quantitative results of long-term future frame prediction from seen / novel viewpoints on PhysInOne.

	PMF $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
TiNeuVox [31]	3.710 / 2.885	21.49 / 15.20	0.633 / 0.452	0.517 / 0.665
DefGS [90]	3.980 / 3.347	22.85 / 17.95	0.833 / 0.598	0.192 / 0.348
TRACE [53]	3.869 / 3.242	22.42 / 17.44	0.756 / 0.599	0.295 / 0.422
FreeGave [54]	3.897 / 3.265	22.57 / 17.75	0.818 / 0.619	0.219 / 0.355
ExtDM [98]	3.363 / -	19.55 / -	0.657 / -	0.771 / -
MAGI-1 [1]	4.086 / -	23.14 / -	0.788 / -	0.364 / -

From Table 3, we can see that current methods demonstrate reasonable future frame prediction for trained viewpoints, but fail to maintain quality under novel viewpoints. This underscores the significant challenge of modeling complex physical motions in 3D space, a gap we expect our

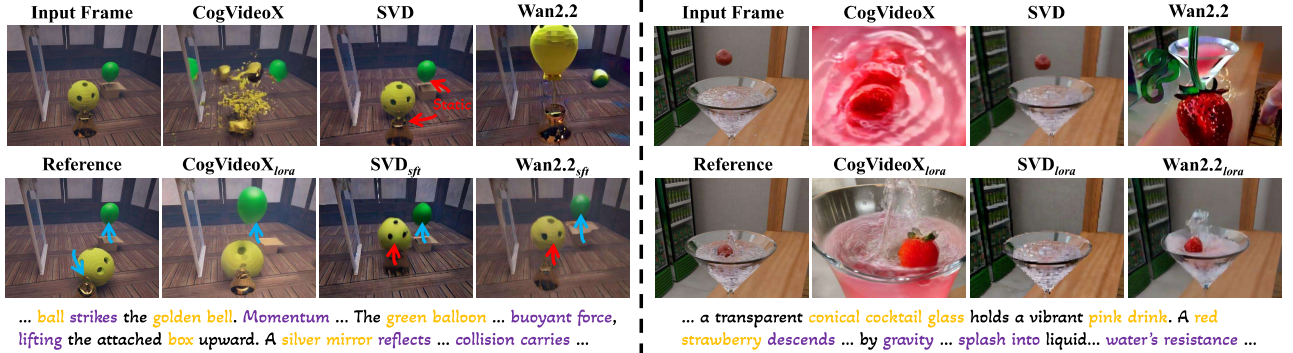


Figure 3. Qualitative examples demonstrating improved physical plausibility in videos generated after fine-tuning on PhysInOne.

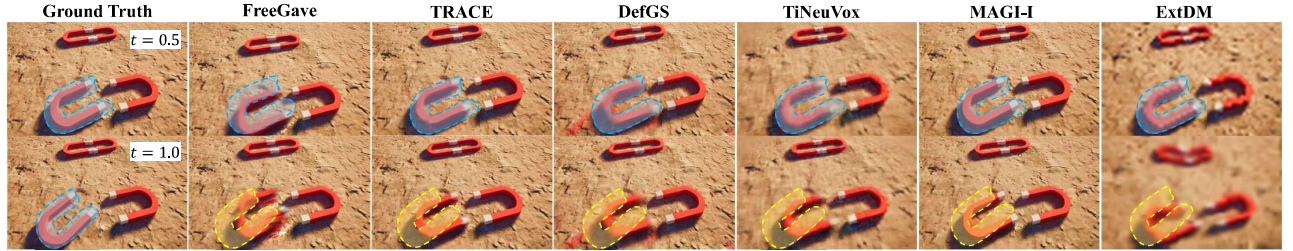


Figure 4. Qualitative examples of long-term future frame prediction by current methods for trained viewpoints.

dataset will inspire more advanced methods to address. Figure 4 shows qualitative results.

4.2.2. Continuous Short-term Future Frame Prediction

In this task, we emulate potential robotic manipulation scenarios by requiring models to continuously predict the next 10 frames in a real time manner, given a stream of continuously observed frames as input. For efficiency, we evaluate four methods **DefGS** [90], **FreeGave** [54], **ExtDM** [98], and **MAGI-1** [1] on the same set of 103 dynamic 3D scenes on both seen and novel viewing angles. Table 4 shows the quantitative results. More details of experiment settings are provided in Appendix 5.8.

Table 4. Quantitative results of short-term future frame prediction from seen / novel viewpoints on PhysInOne.

	PMF $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
DefGS [90]	4.536 / 3.728	26.02 / 20.92	0.861 / 0.739	0.206 / 0.322
FreeGave [54]	4.742 / 3.706	27.09 / 20.80	0.876 / 0.715	0.199 / 0.336
ExtDM [98]	3.774 / -	22.14 / -	0.717 / -	0.715 / -
MAGI-1 [1]	4.696 / -	26.75 / -	0.886 / -	0.116 / -

4.3. Physical Properties Estimation

Estimating the physical properties of objects from visual frames is particularly challenging due to complex dynamics arising from internal forces that vary with object deformation, non-uniform velocity distributions, and spatially heterogeneous external forces. These factors contribute to highly intricate visual appearances. Existing methods are often evaluated on relatively small datasets with limited object diversity, constraining progress in the field. Our collection, encompassing a wide range of objects, including

Table 5. Quantitative results of physical parameters estimation for five types of materials on PhysInOne.

	Elastic Solids			
	$\log_{10}(E)$	ν	$\log_{10}(\gamma_Y)$	v
PAC-NeRF [55]	117.18 ± 68.44	14.26 ± 7.94	25.51 ± 18.34	4.04 ± 1.11
GIC [14]	49.76 ± 8.78	16.35 ± 1.36	-	3.32 ± 1.34
	Plasticine			
	$\log_{10}(E)$	ν	$\log_{10}(\gamma_Y)$	v
PAC-NeRF [55]	68.38 ± 44.14	15.79 ± 11.07	25.51 ± 18.34	3.25 ± 0.23
GIC [14]	178.36 ± 46.68	42.72 ± 25.65	17.11 ± 19.38	3.39 ± 0.87
	Newtonian Fluids			
	$\log_{10}(\mu)$	$\log_{10}(\kappa)$	$\log_{10}(\gamma_Y)$	v
PAC-NeRF [55]	42.64 ± 29.88	287.56 ± 194.98	-	3.11 ± 0.31
GIC [14]	8.78 ± 9.64	70.07 ± 53.44	-	3.28 ± 0.85
	Non-Newtonian Fluids			
	$\log_{10}(\mu)$	$\log_{10}(\kappa)$	$\log_{10}(\gamma_Y)$	v
PAC-NeRF [55]	309.42 ± 235.61	552.89 ± 105.24	339.20 ± 145.99	65.60 ± 71.73
GIC [14]	124.26 ± 98.35	181.87 ± 113.70	28.78 ± 13.89	24.97 ± 15.53
	Granular Substances			
	θ_{fric}	$\log_{10}(\mu)$	$\log_{10}(\kappa)$	v
PAC-NeRF [55]	16.87 ± 27.36	-	-	3.29 ± 0.21
GIC [14]	18.85 ± 16.67	-	-	3.57 ± 0.95

deformable, granular, and liquid types together with complex backgrounds, provides an ideal platform for testing the capabilities of current and future models.

In this section, we evaluate two representative and scene-specific methods **PAC-NeRF** [55] and **GIC** [14] on a randomly selected subset of 20 scenes in our test set, called *test-tiny*. The scenes are distributed across five representative material categories (4 scenes per category):

- **Elastic Solids**: whose dynamics are governed by **Young’s modulus E** and **Poisson’s ratio ν** .
- **Plasticine**: whose dynamics are governed by **Young’s modulus E** , **Poisson’s ratio ν** , and **yield stress γ_Y** .
- **Newtonian Fluids**: whose dynamics are governed by **fluid viscosity μ** and **bulk modulus κ** .



Figure 5. Qualitative resimulation results using estimated physical properties. Both baselines fail to accurately infer properties for complex objects against intricate backgrounds, leading to physically implausible outcomes.

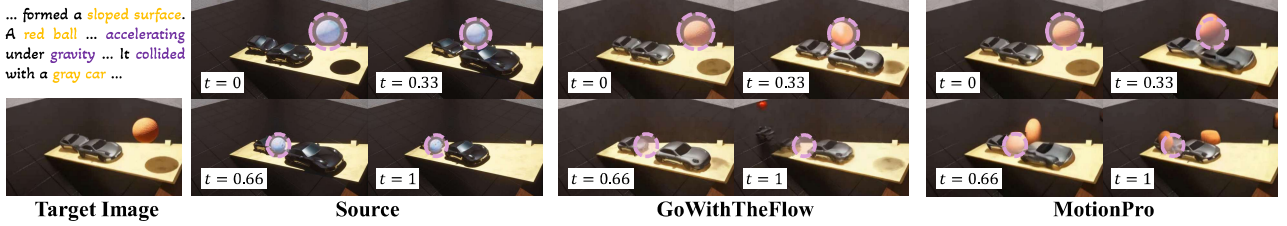


Figure 6. Qualitative motion transfer results from GoWithTheFlow and MotionPro. Generated frames retain visual realism but fail to transfer complex physical motions (e.g., moving cars, falling ball).

- *Non-Newtonian Fluids*: whose dynamics are governed by **yield stress** τ_Y , **shear modulus** μ , **plasticity viscosity** η , and **bulk modulus** κ .
- *Granular Substances*: whose dynamics are governed by **friction angle** θ_{fric} .

Table 6. Quantitative results of resimulation based on estimated physical properties.

	PMF $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
PAC-NeRF [55]	5.617	24.12	0.942	0.086
GIC [14]	5.938	26.90	0.950	0.074

We additionally estimate the initial velocity v of the dynamic object in each test scene. Table 5 compares the accuracy of estimated physical properties against the ground truth provided in PhysInOne. To further validate the learned properties, we resimulate 3D scenes under novel initial conditions (e.g., modified object positions) and quantitatively compare rendered videos against reference videos generated using ground-truth physics under identical novel conditions in Table 6. Figure 5 shows the qualitative results of resimulation. We can see that, while both models infer physically plausible properties, they fall short of accuracy in scenes with complex objects and backgrounds featured by our dataset. This demonstrates how our dataset’s challenging cases provide critical benchmarks for revealing limitations in current physical reasoning capabilities. More details of experiment settings are provided in Appendix 5.9.

4.4. Motion Transfer

Motion transfer seeks to propagate motion dynamics from a source video to a target image, synthesizing a new video that retains the target’s visual attributes while adopting the source’s motion patterns. Current approaches leverage optical flow to encode motion, achieving compelling results on datasets like DAVIS [67]. However, DAVIS primarily

features simple motion patterns, such as a solitary swan gliding across water, with limited physical complexity. This raises a significant question about whether these methods can accurately transfer multiphysics interactions involving multiple objects from one video to another. This capability is essential for advanced applications like film and animation production, and virtual prototyping.

This section evaluates recent methods **MotionPro** [99] and **GoWithTheFlow** [13] using a subset of 273 dynamic 3D scenes (source) from the PhysInOne val set. For quantitative assessment, we generate paired target scenes by replacing source objects with alternative shapes and materials while preserving identical physics activities. New videos rendered from these target scenes enable motion transfer performance evaluation. In Table 7 and Figure 6, both methods maintain high visual fidelity overall, but fail to transfer intricate physical motions accurately, underscoring fundamental challenges in modeling complex physical dynamics, a gap we expect our dataset will motivate future methods to fill. More details of experiment settings are in Appendix 5.10.

Table 7. Quantitative results of transferring physical motions.

	PMF $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
GoWithTheFlow [13]	3.309	18.98	0.691	0.410
MotionPro [99]	3.484	20.28	0.775	0.467

5. Conclusion

We introduce **PhysInOne**, a comprehensive dataset featuring 153,810 dynamic 3D scenes and 2 million videos covering 71 everyday physical phenomena. Experiments across four tasks demonstrate that fine-tuning on PhysInOne significantly improves the physical plausibility of foundation models. However, existing methods still struggle to accurately learn complex multiobject dynamics and physical properties, validating the dataset’s value as a rigorous testbed.

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